EMS611U: Conceptual Design of Aircraft

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Week 5

Aerofoil & Geometry Selection

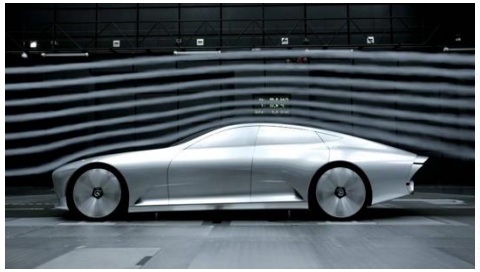
Introduction

What does an aerofoil affect?

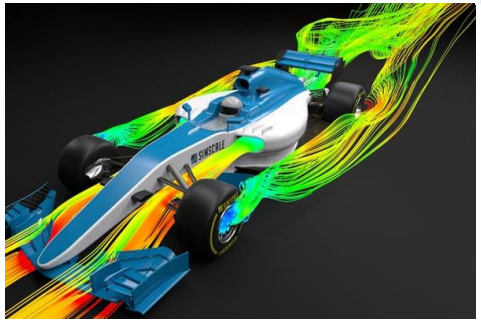
- cruise speed
- takeoff and landing distances
- stall speed
- handling qualities (especially near the stall)
- overall *aerodynamic efficiency* during all phases of flight.

Different operations/missions/aircraft require different geometries.





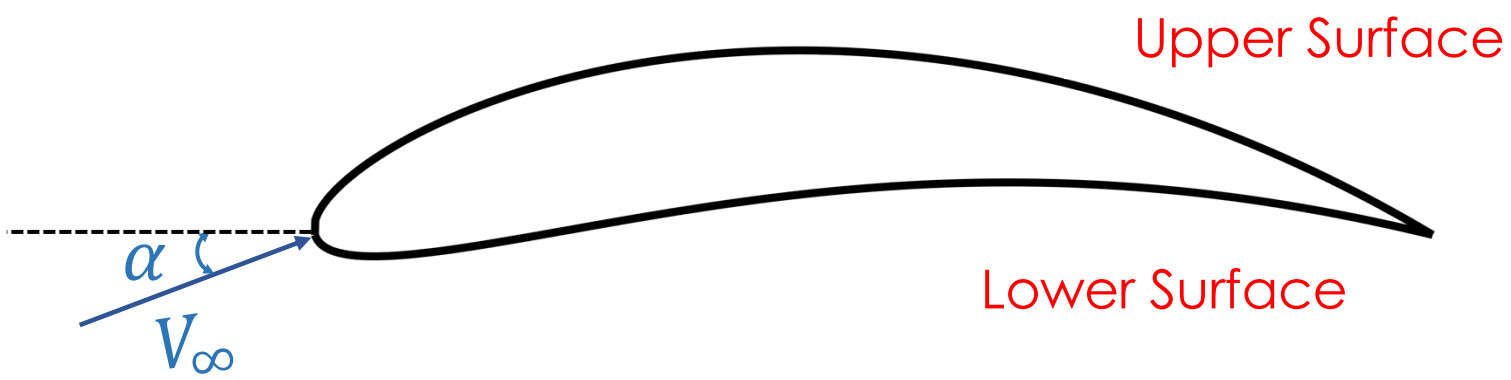








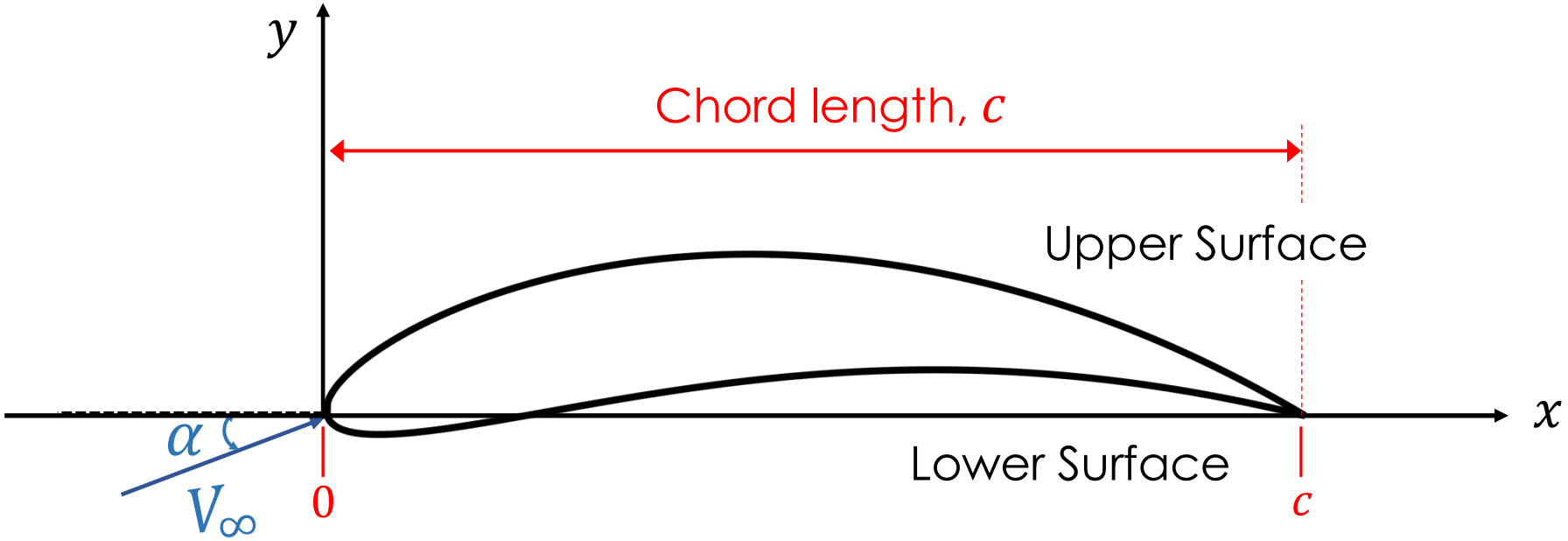
Aerofoil Geometry



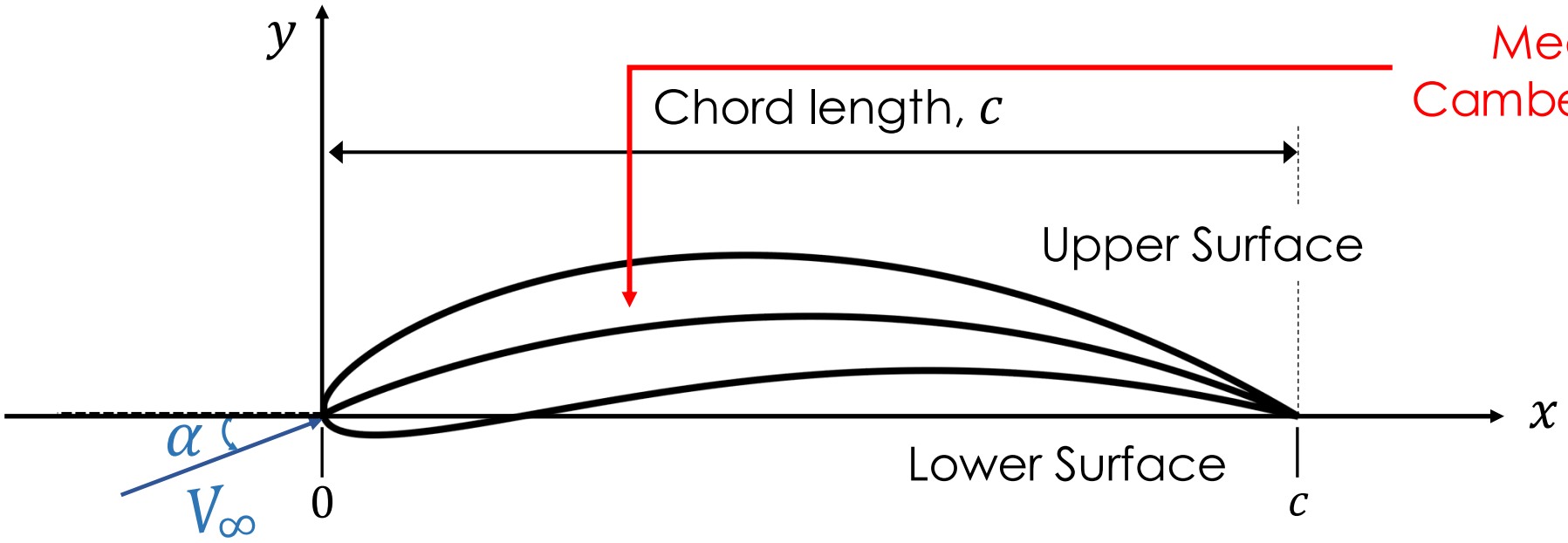
α : Angle of Attack

V_∞: Oncoming Air

Aerofoil Geometry

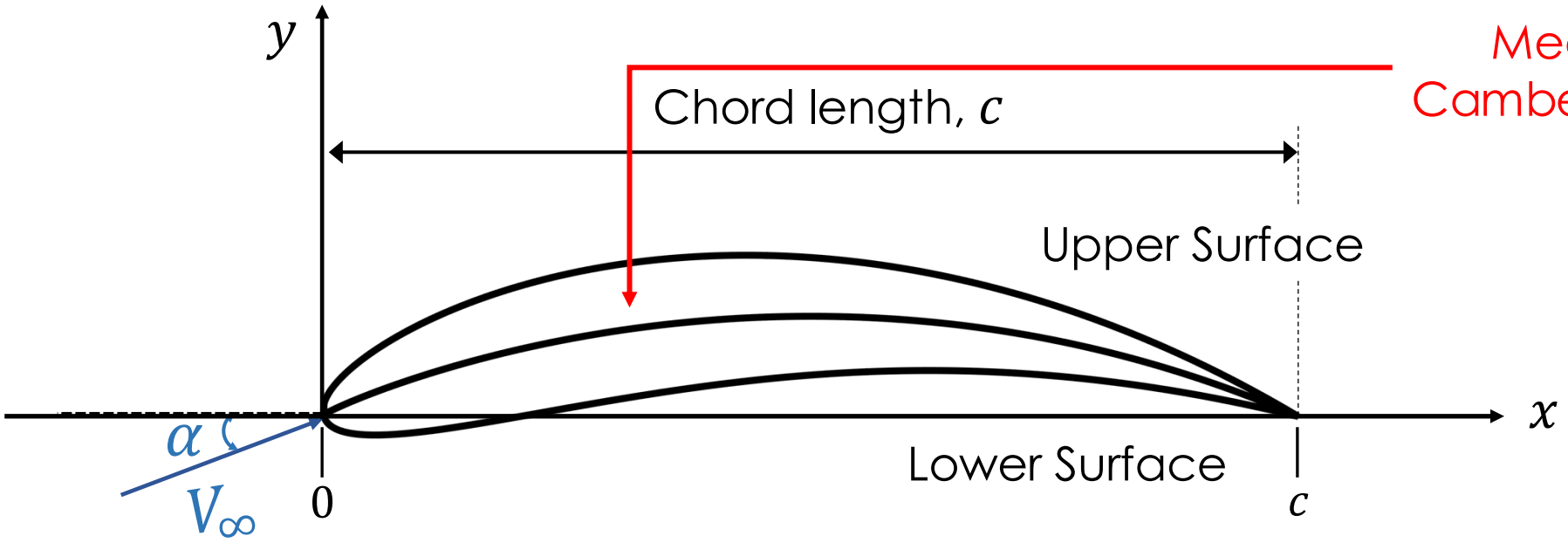


Aerofoil Geometry



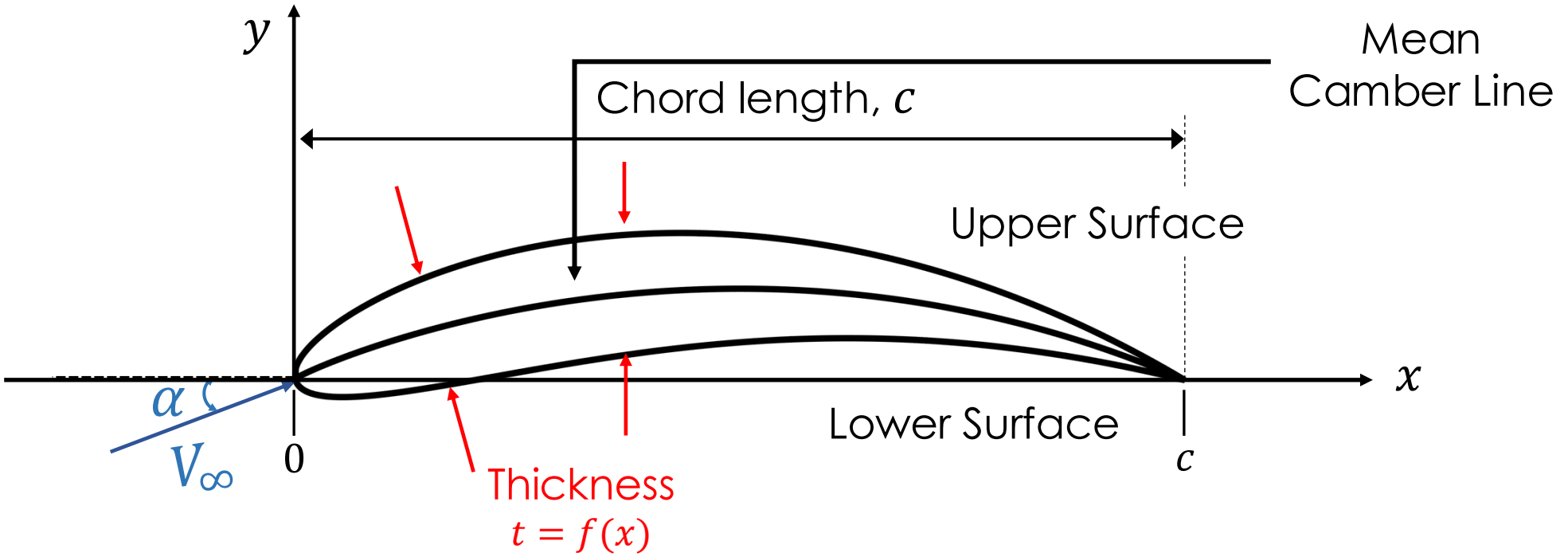
- **Mean Camber Line:** is a line joining the mid points between upper and lower surfaces.

Aerofoil Geometry



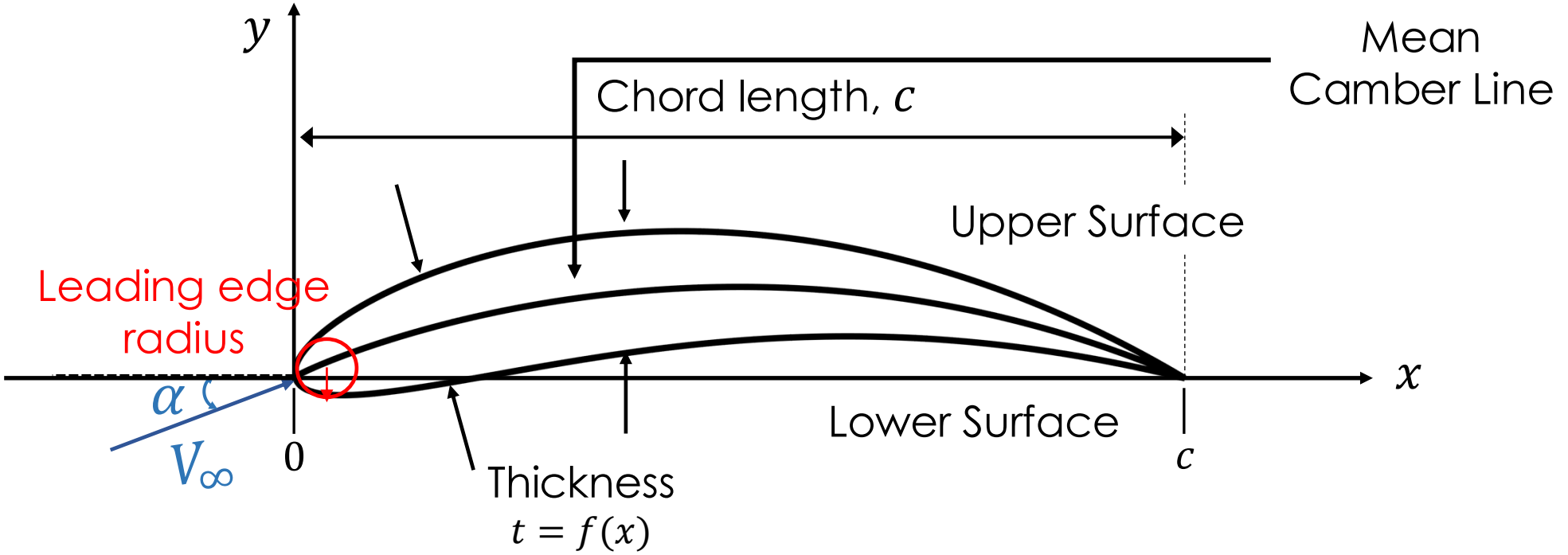
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- **Chord line:** is a straight line that joins the end points of the mean camber line.

Aerofoil Geometry



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- **Airfoil thickness ratio:** t/c

Aerofoil Geometry



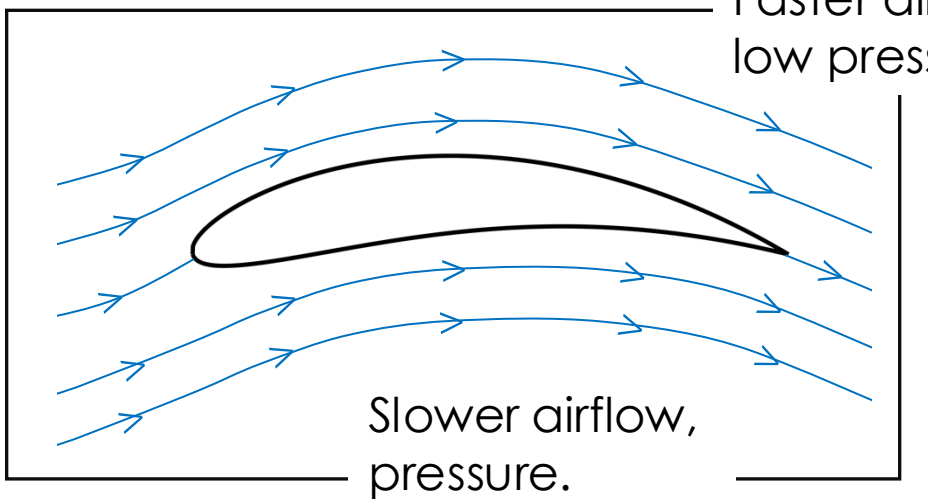
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- **Chord line:** is a straight line that joins the end points of the mean camber line.
- **Thickness distribution:** is the distance from upper and lower surface, measured perpendicular to the mean camber line, and is a function of the distance from the leading edge.
- **Airfoil thickness ratio:** t/c
- The front of the airfoil is defined by **a leading-edge radius**.
  - If supersonic, sharp (or nearly-sharp) leading edge to prevent bow-shock.
- It is very difficult to manufacture a perfectly sharp trailing edge.

Aerofoil Lift and Drag

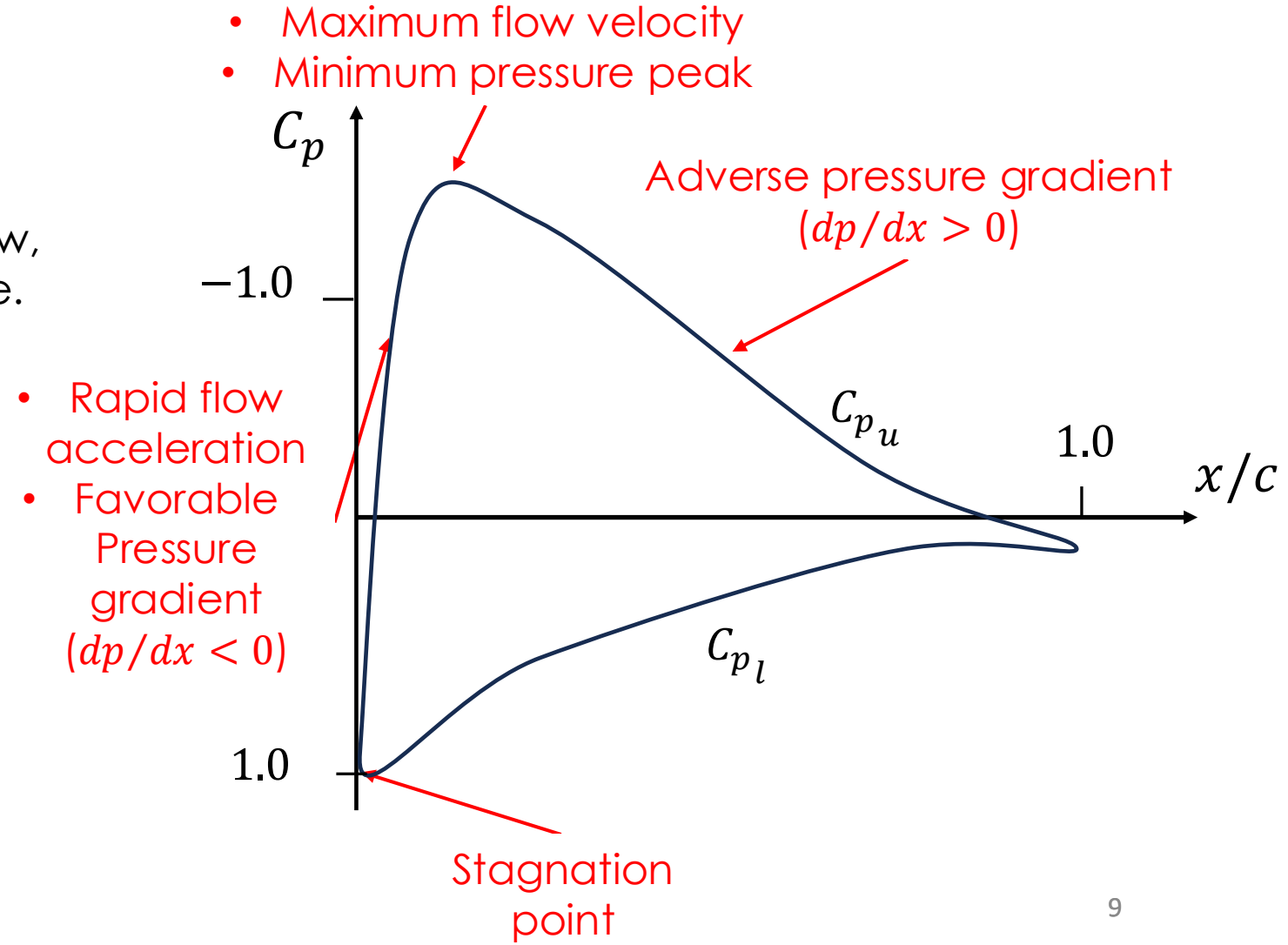
- An aerofoil generates lift by changing the velocity of the air passing over and under itself.
- The aerofoil angle of attack and/or camber causes the air over the top of the airfoil to travel faster than the air beneath it.

Bernoulli's Equation

- $P_1 + \frac{1}{2}\rho V_1^2 = constant$



$$C_p = \frac{P - P_\infty}{\frac{1}{2}\rho V_\infty^2}$$

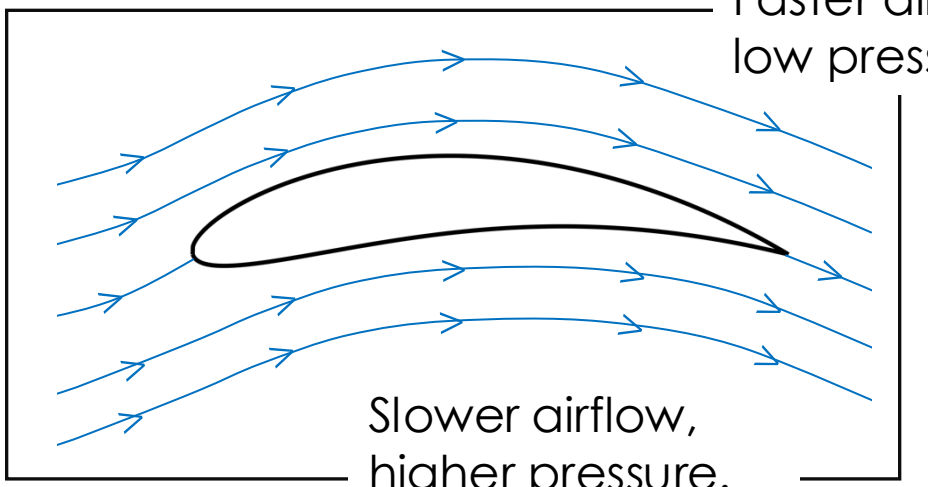


Aerofoil Lift and Drag

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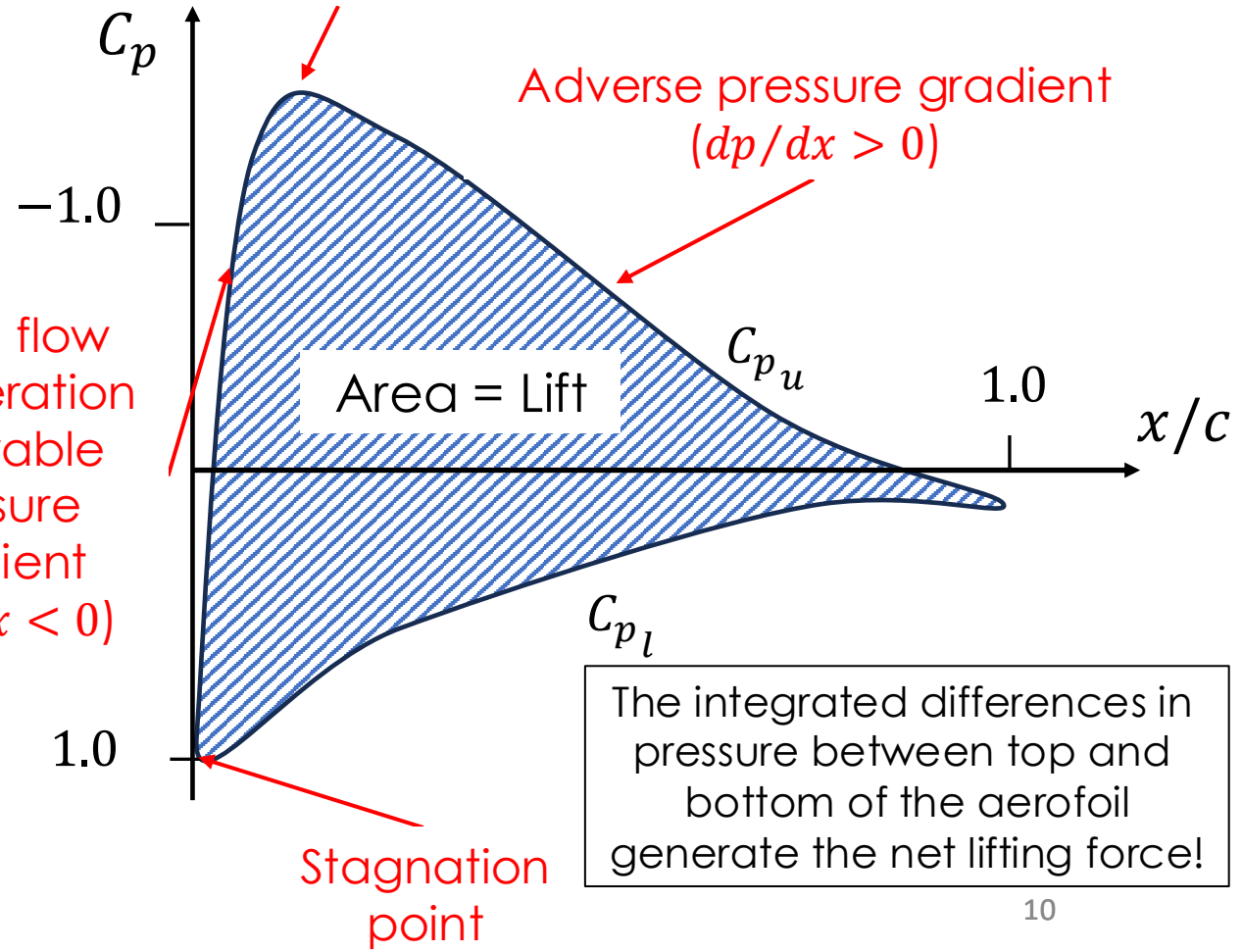
Bernoulli's Equation

- $P_1 + \frac{1}{2}\rho V_1^2 = constant$



$$C_p = \frac{P - P_\infty}{\frac{1}{2}\rho V_\infty^2}$$

- Maximum flow velocity
- Minimum pressure peak

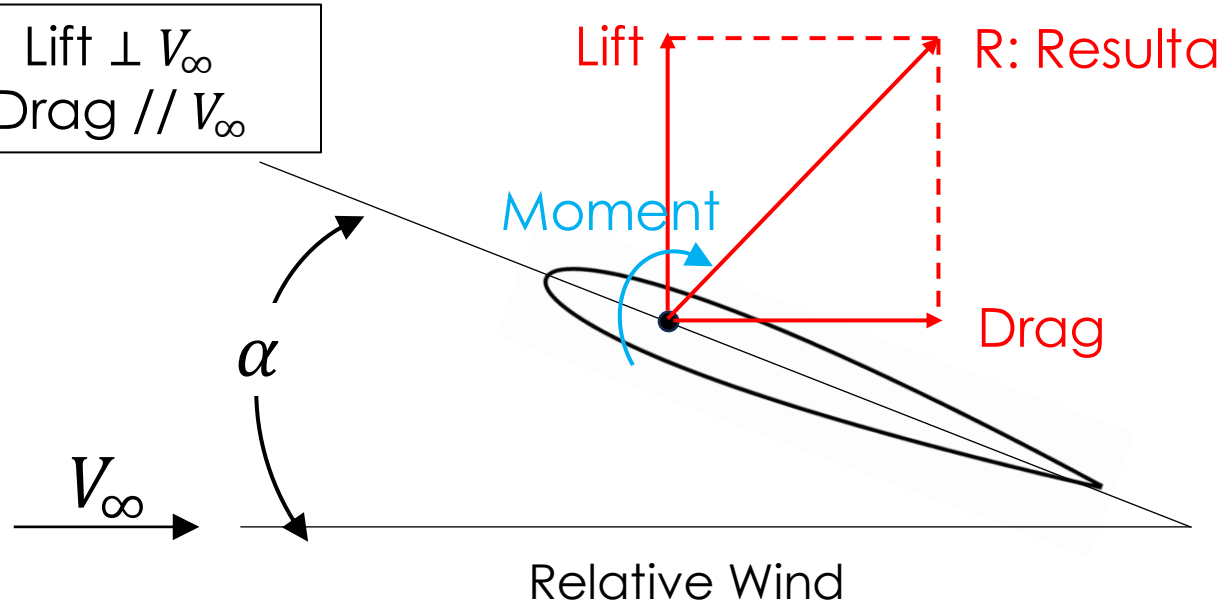


- Rapid flow acceleration
- Favorable Pressure gradient $(dp/dx < 0)$

The integrated differences in pressure between top and bottom of the aerofoil generate the net lifting force!

Aerofoil Lift and Drag

Lift $\perp V_\infty$
Drag // V_∞



Lift, drag, moment on an airfoil:

- l =Lift/unitspan
- d =drag/unitspan
- m =Moment/unitspan

Dimensional analysis gives:

Forces and Moments are a function of:

- α: Angle of attack
- V_∞: Freestream velocity
- ρ_∞: Freestream density
- c: Airfoil chord length
- μ_∞: Freestream viscosity
- a_∞: Freestream sonic speed
- q_∞: Freestream dynamic pressure

$$C_l = \frac{l}{\frac{1}{2}\rho V_\infty^2 c}$$

$$C_d = \frac{d}{\frac{1}{2}\rho V_\infty^2 c}$$

$$C_m = \frac{m}{\frac{1}{2}\rho V_\infty^2 c^2}$$

$$f(\alpha, Re, M_\infty)$$

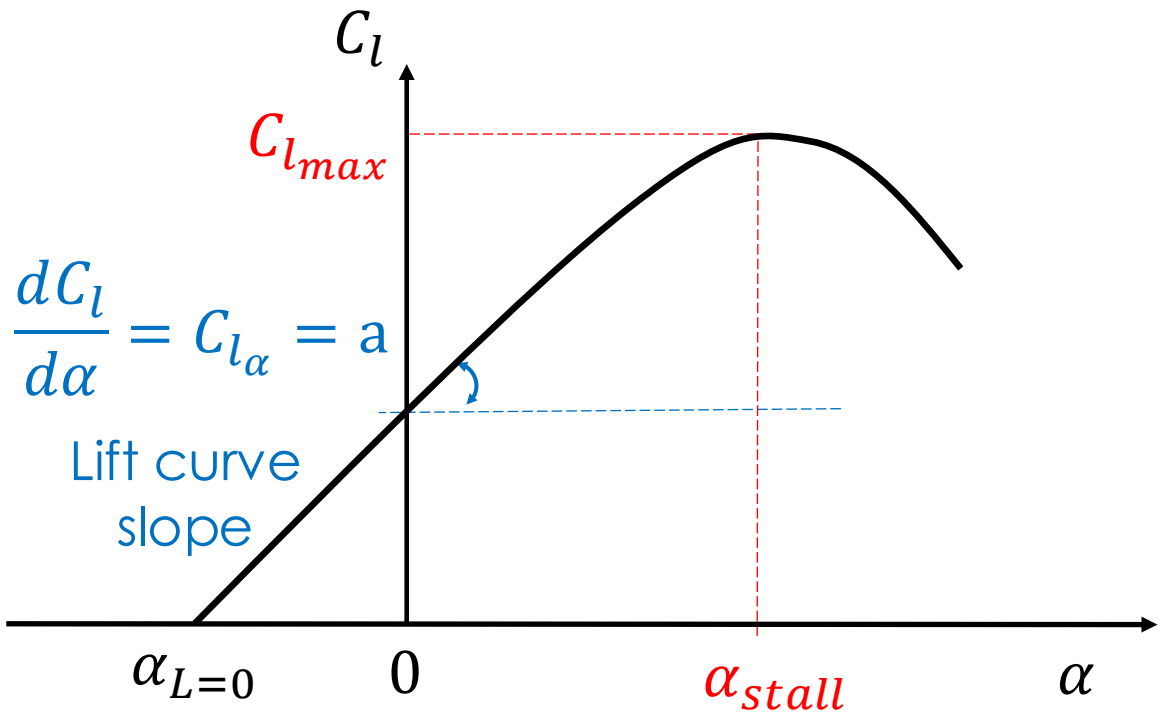
$$Re = \frac{\rho_\infty V_\infty c}{\mu_\infty}$$

$$M_\infty = \frac{V_\infty}{a_\infty}$$

Important Aerofoil Characteristics

Lift Curve:

- C_l vs. α



Zero-lift angle of attack

For linear-part of the lift curve:

$$C_l = C_{l_\alpha}(\alpha - \alpha_{L=0}) = a(\alpha - \alpha_{L=0})$$

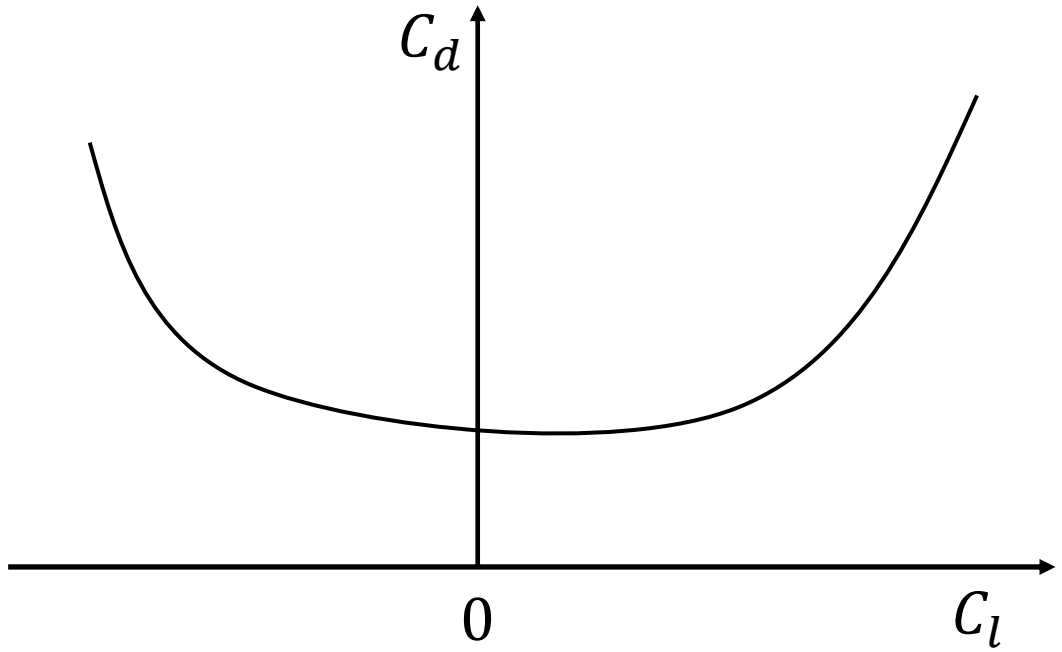
$$C_{l_\alpha} = a\ [1/\text{rad}]$$

- Theoretical value for the lift curve slope ($C_{l_\alpha} = a$) is 2π
- If zero-lift angle of attack ($\alpha_{L=0} = 0$), we have a *symmetric aerofoil.*

Important Aerofoil Characteristics

Drag Polar:

- C_l vs. C_d

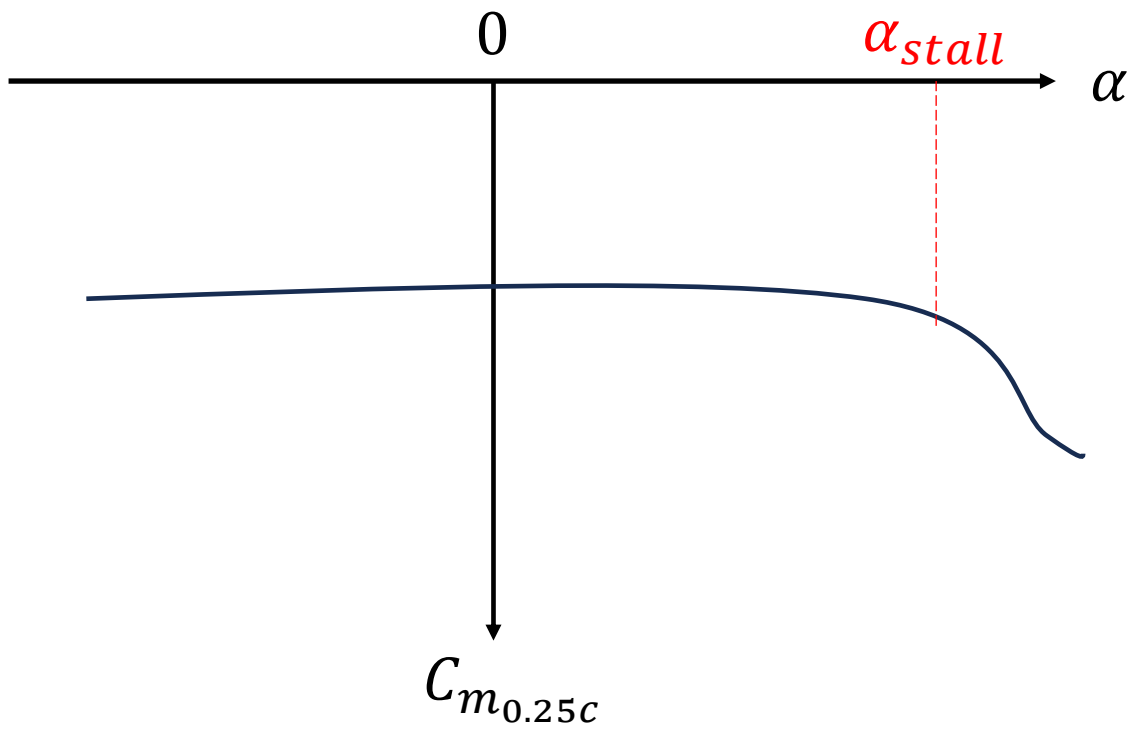


- A measure of aerodynamic efficiency: $\boldsymbol{C_l/C_d}$
- Maximum of this parameter is obtained by drawing the tangent from origin to the curve.

Important Aerofoil Characteristics

Pitching Moment Curve:

- C_m vs. α



- Magnitude of the pitching moment depends on the moment reference center.
- $C_{m_{0.25c}}$ is about to quarter chord point as identified in the subscript. The pitching moment is almost independent of angle of attack about the quarter-chord point for most airfoils at subsonic speeds The (aerodynamic center is generally the quarter-chord).

Reynolds Number Effects

Forces and Moments are a function of:

- α: Angle of attack
- Re_∞: Reynolds Number
- M_∞: Mach Number

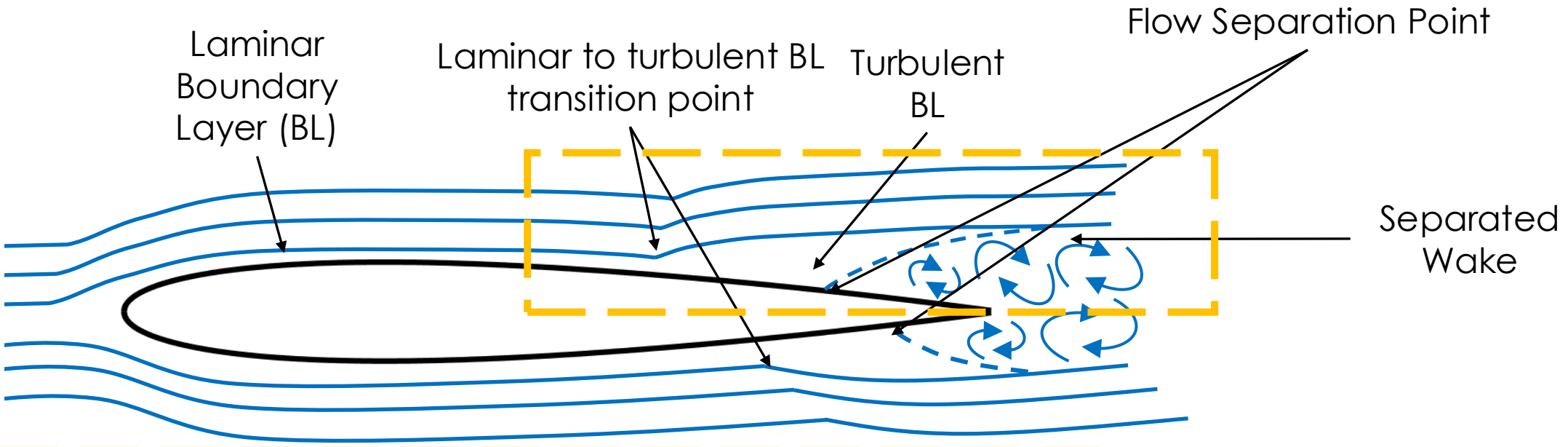
$$Re = \frac{\rho_\infty V_\infty c}{\mu_\infty}$$

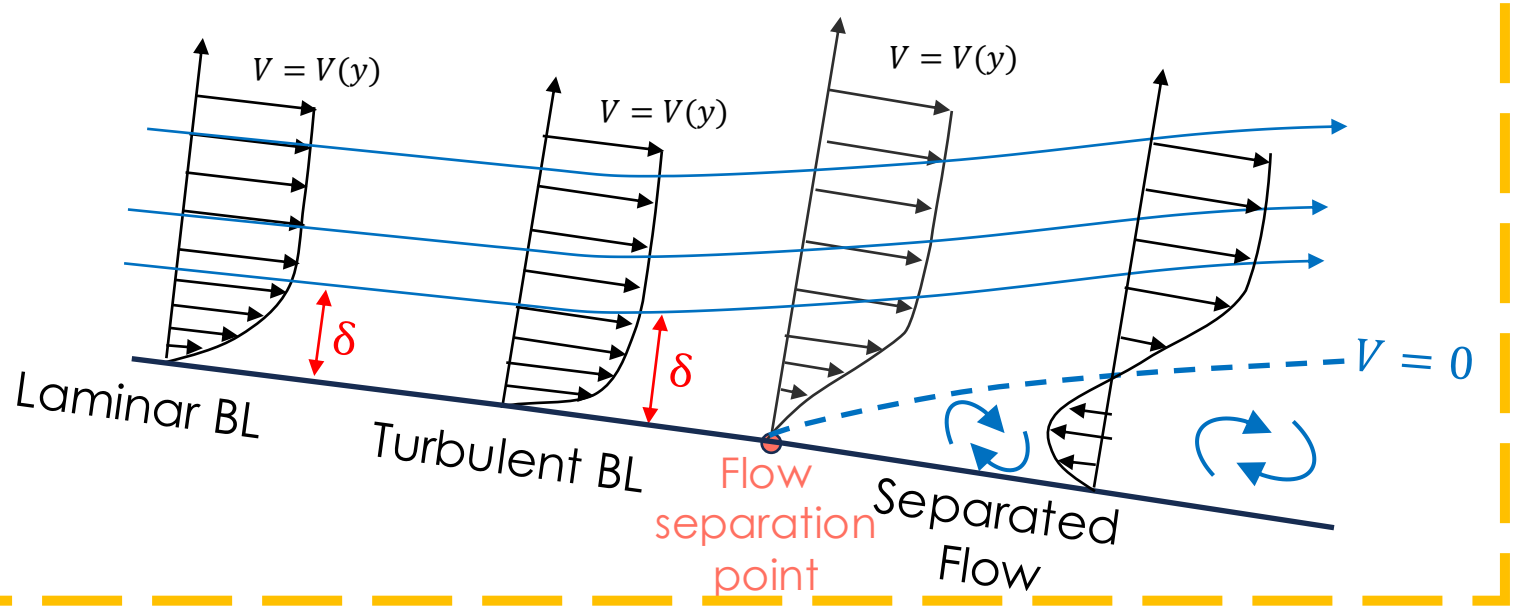
$\rho_\infty V_\infty c$ } Inertia

μ_∞ } Viscous

- *Airfoil characteristics are strongly affected by the "Reynolds number" at which they are operating.*
- *The Reynolds number influences whether the flow will be laminar or turbulent and whether flow separation will occur.*
- A typical aircraft wing operates at a Reynolds number of about ten milllion.

Reynolds Number Effects

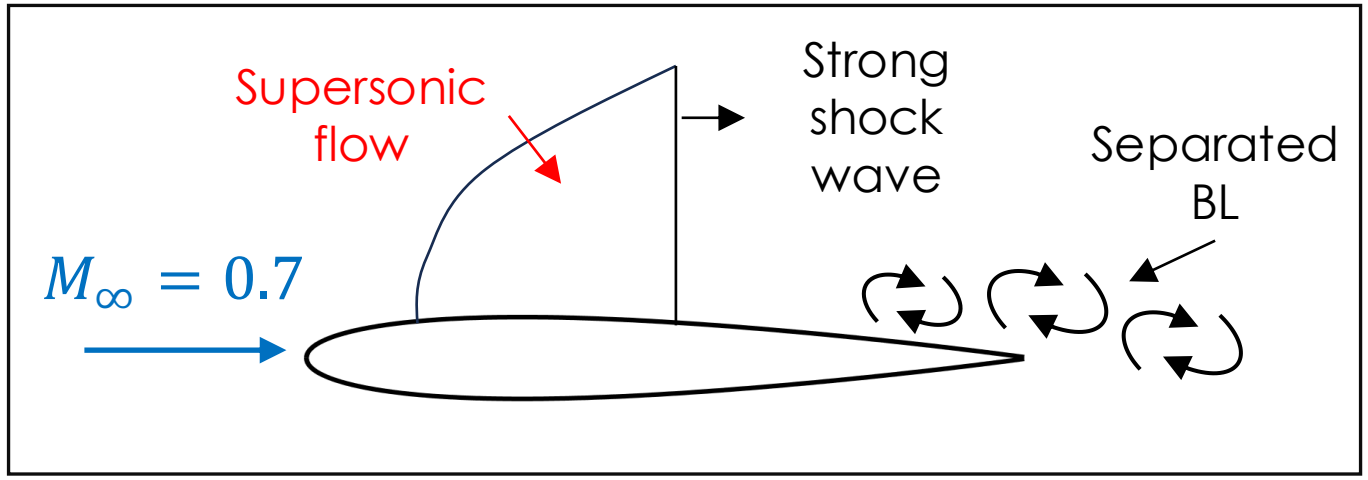


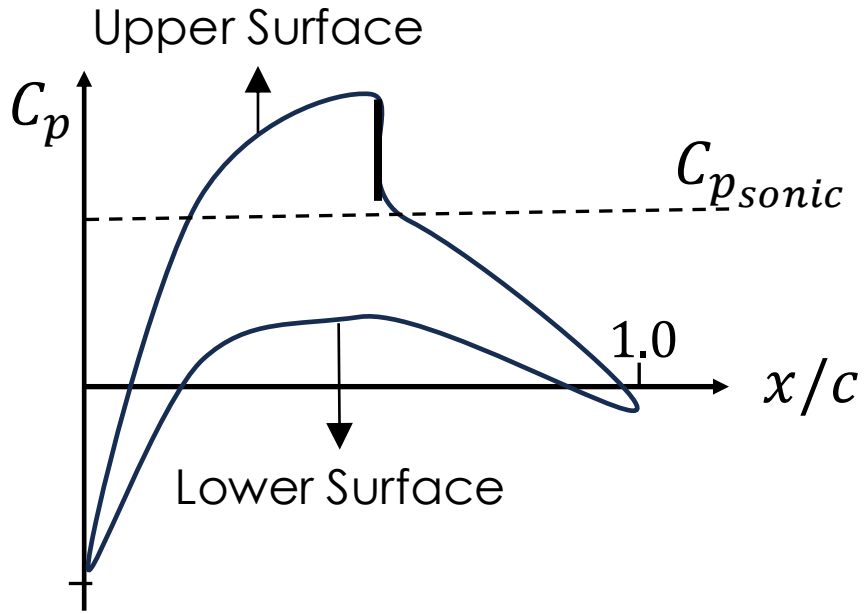


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Mach Number Effects

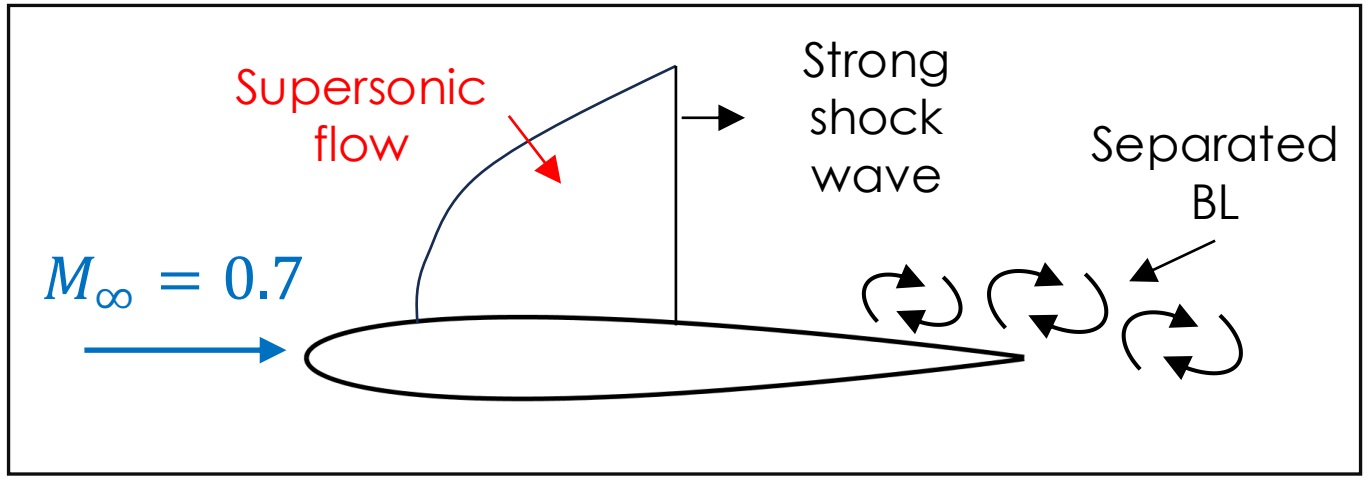
- Due to the variation of local velocity over the curved surface of an airfoil, the local Mach number of the flow can vary considerably from that of the free stream.
- By increasing the freestream Mach number at a given angle of attack, eventually a freestream Mach number is reached for which sonic speed occurs somewhere on the airfoil surface. That free stream Mach number from which this first occurs is defined as the ***critical Mach number.***

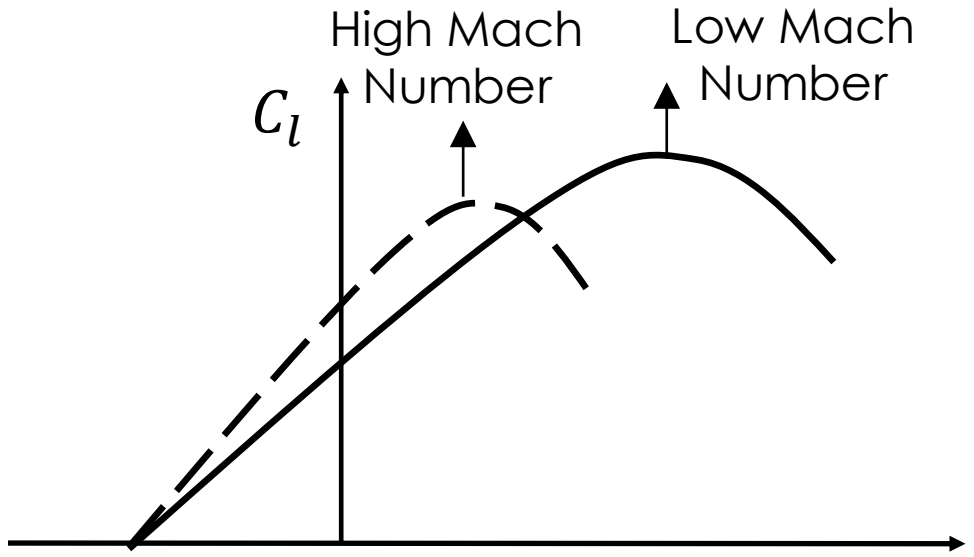




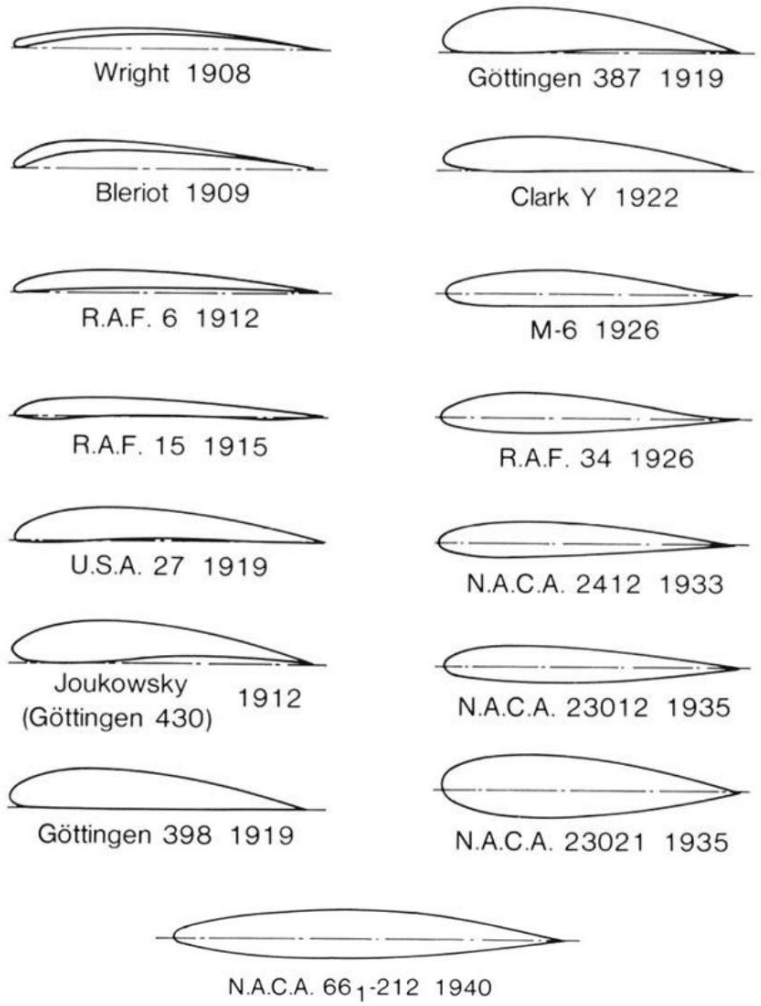
Mach Number Effects

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Aerofoil Families



- Early aerofoils are developed mostly by trial and error.
- By the1920s, a wide variety of aerofoils appeared, and consolidation was needed.
- Since the 1930s, **NACA** generated families of aerofoils benefiting from what was available in the market and beyond. It presented the aerofoil geometries and test results in a systematic manner, grouping them into family series.
- NACA aerofoil family is listed in with well-calibrated wind-tunnel results and many fine aircraft have used the NACA series of aerofoils.

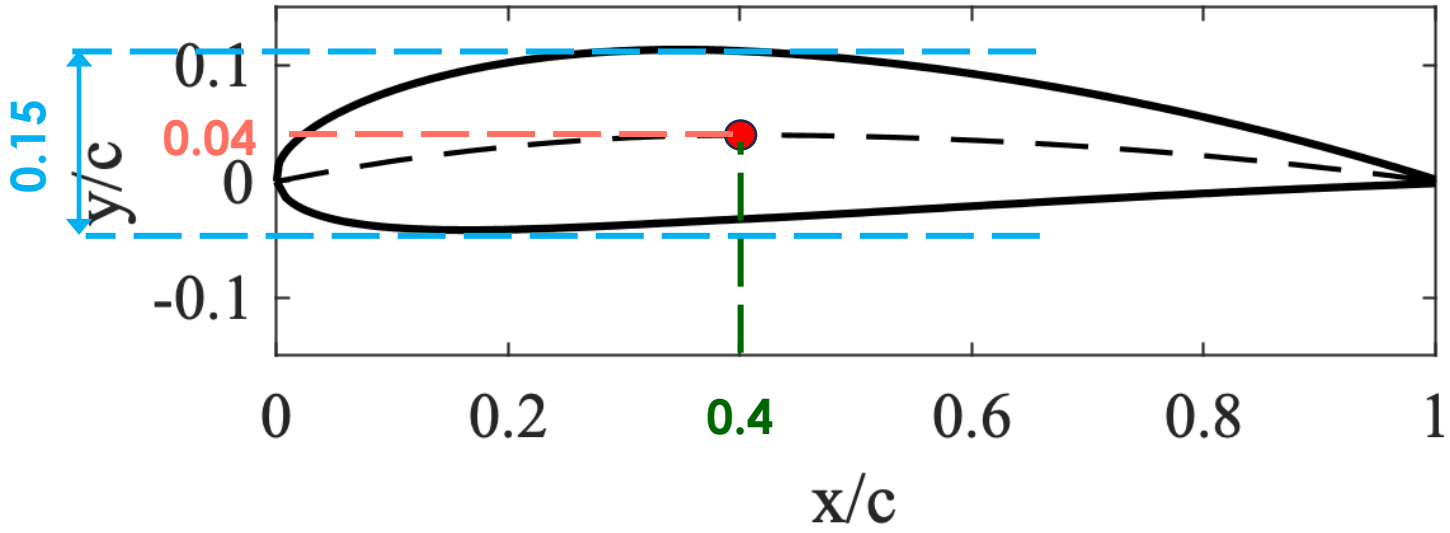
NACA: National Advisory Committee for Aeronautics

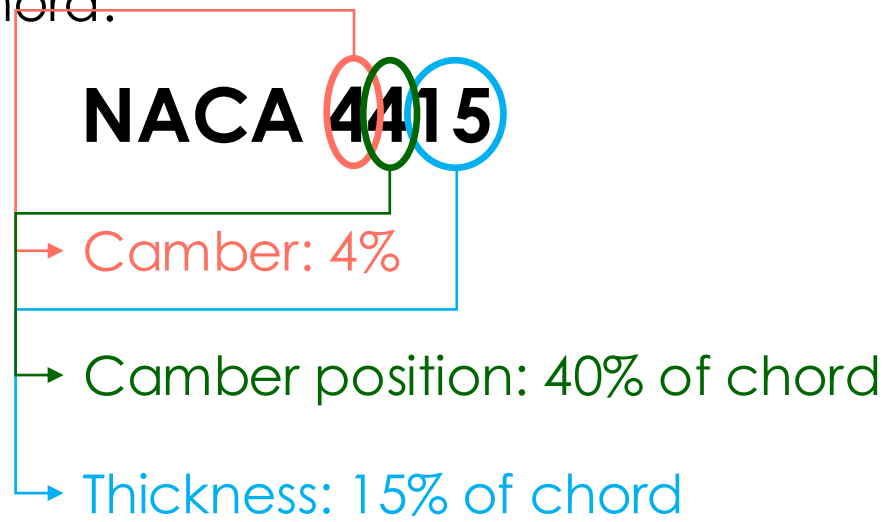
- A predecessor to NASA.

NACA Four-Digit Aerofoil:

- Each of the four digits of the nomenclature represents a geometrical property:

First Digit: Defines the maximum camber in percentage of the chord.
Second Digit: Location of maximum camber in 1/10 of chord.
Last Two Digits: Maximum thickness in percentage of chord.



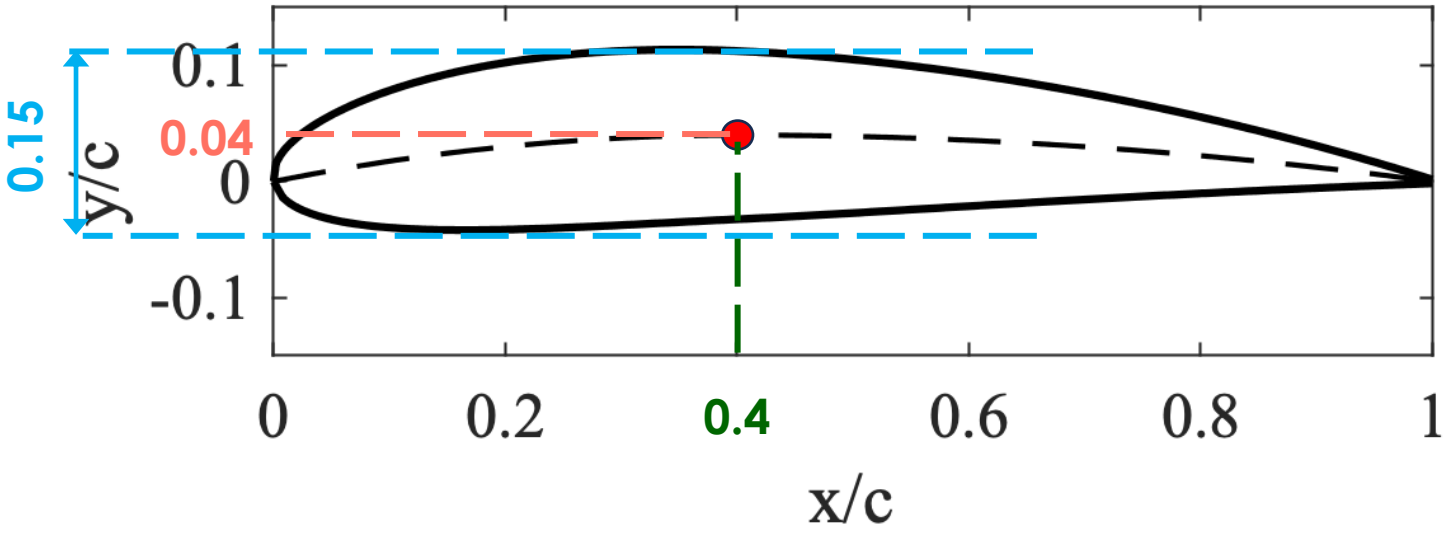


Aerofoil Families – NACA Aerofoils

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NACA 4415

- Camber: 4%
- Camber position: 40% of chord
- Thickness: 15% of chord

NACA 0012

- Camber: 0%
- Camber position: 0% of chord
- Thickness: 12% of chord

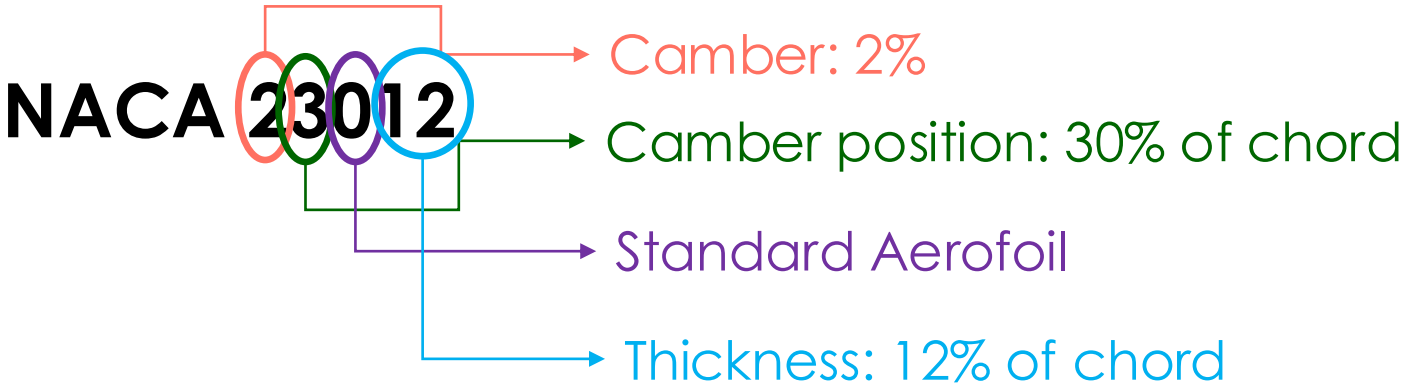
- NACA00: Uncambered aerofoil.
- While rarely used for wing design today, the **uncambered four-digit airfoils** are still commonly used for **tail surfaces of subsonic aircraft.**

Aerofoil Families – NACA Aerofoils

NACA Five-Digit Aerofoil:

- The NACA five-digit airfoils were developed to allow shifting the position of maximum camber forward for greater maximum lift.

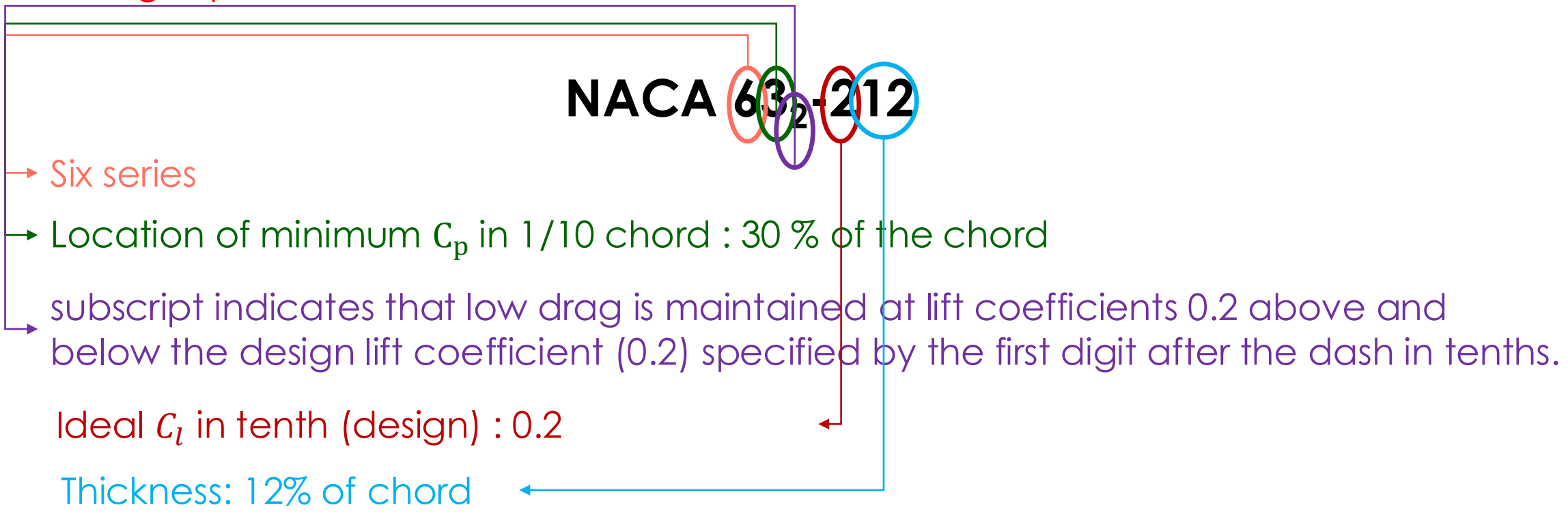
First Digit: Defines the maximum camber in percentage of the chord.
Second Digit: Location of maximum camber in 1/10 of chord.
Third Digit: "0" means a standard 5-digit aerofoil; "1" means a reflexed camber.
Last Two Digits: Maximum thickness in percentage of chord.



Aerofoil Families – NACA Aerofoils

NACA Six-Digit Aerofoil:

- The state-of-the-art for a good aerofoil often follows reverse engineering.
- The six-digit series aerofoils were generated from a more or less prescribed pressure distribution and were designed to achieve some laminar flow, *and hence reduced drag.*
- This was achieved by placing the maximum thickness far back from the LE.
- Their low-speed characteristics behave like the four- and five-digit series but show much better high-speed characteristics.



Muddiest Point Quiz

1. Identify one thing (concept) you learnt from this class.

2. Identify one thing (concept) you did not quite understand from this class.